

Eastern Washington University
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Department of Mathematics

MATH 241: Multivariable Calculus

Dr. Nievergelt

Summer Quarter 2025 - Section 025

July 31st, 2025

Final Project "As Flat As Possible"

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Problem 1

Show that in two dimensions equations (3.5) and (3.6) on page 500 of [6] are mutually equivalent.

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0, \quad \mathbf{x} \in \Omega \quad (3.5)$$

$$(1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy} = 0 \quad (3.6)$$

Solution: We first simplify equation (3.5) in the following manner:

$$\begin{aligned} \nabla \cdot \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) &= 0 \\ f &= \sqrt{1 + |\nabla u|^2} \quad \text{Substitution for simplification} \\ &= \nabla \cdot \left(\frac{\nabla u}{f} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{u_x}{f} \right) + \frac{\partial}{\partial y} \left(\frac{u_y}{f} \right) \\ &= \frac{f \cdot u_{xx} - u_x \cdot \frac{\partial f}{\partial x}}{f^2} + \frac{f \cdot u_{yy} - u_y \cdot \frac{\partial f}{\partial y}}{f^2} \quad (\text{Quotient Rule}) \\ &= \frac{f \cdot u_{xx} - u_x \cdot \frac{\partial f}{\partial x} + f \cdot u_{yy} - u_y \cdot \frac{\partial f}{\partial y}}{f^2} \end{aligned}$$

Now we can focus in computing $\frac{\partial f}{\partial x}$:

$$\begin{aligned} &\frac{\partial}{\partial x} \left((1 + u_x^2 + u_y^2)^{\frac{1}{2}} \right) \\ &= \frac{1}{2} \cdot (1 + u_x^2 + u_y^2)^{-\frac{1}{2}} \cdot \frac{\partial}{\partial x} (1 + u_x^2 + u_y^2) \\ &= \frac{1}{2} \cdot (1 + u_x^2 + u_y^2)^{-\frac{1}{2}} \cdot (2u_x u_{xx} + 2u_y u_{yx}) \\ &= \frac{u_x u_{xx} + u_y u_{yx}}{\sqrt{1 + u_x^2 + u_y^2}} \end{aligned}$$

Now we can focus in computing $\frac{\partial f}{\partial y}$:

$$\begin{aligned} &\frac{\partial}{\partial y} \left((1 + u_x^2 + u_y^2)^{\frac{1}{2}} \right) \\ &= \frac{1}{2} \cdot (1 + u_x^2 + u_y^2)^{-\frac{1}{2}} \cdot \frac{\partial}{\partial y} (1 + u_x^2 + u_y^2) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \cdot (1 + u_x^2 + u_y^2)^{-\frac{1}{2}} \cdot (2u_x u_{xy} + 2u_y u_{yy}) \\
&= \frac{u_x u_{xy} + u_y u_{yy}}{\sqrt{1 + u_x^2 + u_y^2}}
\end{aligned}$$

Now we can compute and simplify equation (3.5) as follows:

$$\begin{aligned}
&= \left(\frac{u_{xx}}{\sqrt{1 + u_x^2 + u_y^2}} - u_x \cdot \left(\frac{u_x u_{xx} + u_y u_{yx}}{\sqrt{1 + u_x^2 + u_y^2}} \right) + \frac{u_{yy}}{\sqrt{1 + u_x^2 + u_y^2}} - u_y \cdot \left(\frac{u_x u_{xy} + u_y u_{yy}}{\sqrt{1 + u_x^2 + u_y^2}} \right) \right) \cdot \frac{1}{1 + u_x^2 + u_y^2} \\
&= \frac{u_{xx} - u_x^2 u_{xx} - u_y u_x u_{yx} + u_{yy} - u_x u_y u_{xy} - u_y^2 u_{yy}}{\sqrt{1 + u_x^2 + u_y^2}} \cdot \frac{1}{1 + u_x^2 + u_y^2} \\
&= \frac{u_{xx} + u_y^2 u_{xx} - 2u_x u_y u_{xy} + u_{yy} + u_x^2 u_{yy}}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \quad (\text{By Clairut's Theorem}) \\
&= \frac{(1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy}}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} = 0
\end{aligned}$$

Now since $u_x^2 + u_y^2 \geq 0$ as both partial derivatives are being squared, then adding 1 leads to $1 + u_x^2 + u_y^2 \geq 1$. Therefore, we can simplify our expression by multiplying both sides of the equation by our denominator $(u_x^2 + u_y^2 + 1)^{\frac{3}{2}}$ as it is not equal to zero:

$$\begin{aligned}
&(1 + u_x^2 + u_y^2)^{\frac{3}{2}} \cdot \frac{(1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy}}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} = 0 \cdot (1 + u_x^2 + u_y^2)^{\frac{3}{2}} \\
&(1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy} = 0
\end{aligned}$$

And this expression is exactly equation (3.6). Therefore in two dimensions,

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = (1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy} = 0$$

Problem 2

Show how to derive equation (3.5) from (3.4) on page 500 of [6] using the method on pages 498–499 of [6].

$$SA[u] = \int_{\Omega} \sqrt{1 + |\nabla u|^2} dy dx \quad (3.4)$$

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0, \quad \mathbf{x} \in \Omega \quad (3.5)$$

Solution: Let $\phi = 0$ on $\partial\Omega$, be a function then $SA[u + \phi]$ also agrees with our function g on the boundary of Ω . And since u minimizes $SA[u]$ the following is true:

$$SA[u] \leq SA[u + \phi]$$

Now we can generalize this with a function $i_{\phi}(\epsilon)$ where ϵ is a nonnegative scalar such that:

$$i_{\phi}(\epsilon) = SA[u + \epsilon\phi]$$

Since u minimizes $SA[u]$ then $i_{\phi}(\epsilon)$ has a minimum at $\epsilon = 0$ or $i'_{\phi}(0) = 0$. Now we can compute this derivative as follows:

$$\begin{aligned} i'_{\phi}(0) &= \lim_{h \rightarrow 0} \frac{i_{\phi}(h) - i_{\phi}(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} (SA[u + h\phi] - SA[u]) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} \sqrt{1 + |\nabla u + h\nabla\phi|^2} dy dx - \int_{\Omega} \sqrt{1 + |\nabla u|^2} dy dx \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} \sqrt{1 + |\nabla u|^2 + 2h\nabla u \nabla\phi + h^2|\nabla\phi|^2} dy dx - \int_{\Omega} \sqrt{1 + |\nabla u|^2} dy dx \right) \\ a &= \nabla u, \quad b = \nabla\phi \quad c = \nabla u \nabla\phi \quad A = |\nabla u|^2 \quad B = |\nabla\phi|^2 \quad (\text{Substitution for simplification}) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} \sqrt{1 + A + 2hc + h^2B} dy dx - \int_{\Omega} \sqrt{1 + A} dy dx \right) \\ &\text{Use a Taylor expansion on } \sqrt{1 + A + 2hc + h^2B} : \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} \sqrt{1 + A} + \frac{ch}{\sqrt{1 + A}} + O(h^2) dy dx - \int_{\Omega} \sqrt{1 + A} dy dx \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(h \int_{\Omega} \frac{c}{\sqrt{1 + A}} dy dx + \int_{\Omega} O(h^2) dy dx \right) \end{aligned}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{1}{h} \left(h \int_{\Omega} \frac{c}{\sqrt{1+A}} dy dx + h \int_{\Omega} O(h) dy dx \right) \\
&= \lim_{h \rightarrow 0} \left(\int_{\Omega} \frac{c}{\sqrt{1+A}} dy dx + \int_{\Omega} O(h) dy dx \right) \\
&= \int_{\Omega} \frac{c}{\sqrt{1+A}} dy dx \\
&= \int_{\Omega} \frac{\nabla u \nabla \phi}{\sqrt{1+|\nabla u|^2}} dy dx
\end{aligned}$$

Now we integrate by parts using Green's first identity :

$$f = \frac{1}{\sqrt{1+|\nabla u|^2}} \quad \text{Substitution for simplification}$$

$$\int_{\Omega} f \nabla u \cdot \nabla \phi dy dx = \int_{\partial \Omega} \phi f \nabla u \cdot n ds - \int_{\Omega} \phi \nabla \cdot (f \nabla u) dy dx$$

Since we defined $\phi = 0$ on $\partial \Omega$ then $\int_{\partial \Omega} \phi f \nabla u \cdot n ds = 0$.

$$\int_{\Omega} f \nabla u \cdot \nabla \phi dy dx = - \int_{\Omega} \phi \nabla \cdot (f \nabla u) dy dx$$

$$\int_{\Omega} f \nabla u \cdot \nabla \phi dy dx = - \int_{\Omega} \phi \operatorname{div} \left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}} \right) dy dx$$

Now for this integral to equal 0 for any choice of ϕ then $\operatorname{div} \left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}} \right) = 0$

Therefore, we have arrived the equation (3.5) from (3.4):

$$\boxed{\operatorname{div} \left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}} \right) = 0}$$

Problem 3

Show that the calculations on pages 494–495 of [6] also hold for balls and spheres in three dimensions.

Suppose for every $\mathbf{x}_0$ in the domain of u the value $u(\mathbf{x}_0)$ is the average of the values of u on a sphere centered at $\mathbf{x}_0$:

$$u(\mathbf{x}_0) = \frac{1}{4\pi r^2} \int_{\partial B(\mathbf{x}_0, r)} u(x_0, y_0, z_0) dS$$

Change to Spherical coordinates:

$$\begin{aligned}
&= \frac{1}{4\pi r^2} \int_0^{2\pi} \int_0^\pi u(x_0 + r \sin(\phi) \cos(\theta), y_0 + r \sin(\phi) \sin(\theta), z_0 + r \cos(\phi)) r^2 \sin(\phi) d\phi d\theta \\
&= \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi u(x_0 + r \sin(\phi) \cos(\theta), y_0 + r \sin(\phi) \sin(\theta), z_0 + r \cos(\phi)) \sin(\phi) d\phi d\theta
\end{aligned}$$

Now differentiate with respect to r :

$$\begin{aligned}
\frac{d}{dr}(u(\mathbf{x}_0)) &= \int_0^{2\pi} \int_0^\pi \frac{d}{dr} (u(x_0 + r \sin(\phi) \cos(\theta), y_0 + r \sin(\phi) \sin(\theta), z_0 + r \cos(\phi)) \sin(\phi)) d\phi d\theta \\
0 &= \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\partial u}{\partial x} \cdot (\sin(\phi) \cos(\theta)) + \frac{\partial u}{\partial y} \cdot (\sin(\phi) \sin(\theta)) + \frac{\partial u}{\partial z} \cdot (\cos(\phi)) d\phi d\theta
\end{aligned}$$

Where u is evaluated at

Now we recognize $(\sin(\phi) \cos(\theta), \sin(\phi) \sin(\theta), \cos(\phi))$ as the unit normal vector and $r^2 d\phi d\theta$ as a

$$\begin{aligned}
0 &= \frac{1}{4\pi r^2} \int_0^{2\pi} \int_0^\pi \nabla u \cdot \mathbf{n} dS \\
0 &= \frac{1}{4\pi r^2} \int_{\partial B(\mathbf{x}_0, r)} \nabla u \cdot \mathbf{n} dS
\end{aligned}$$

Now we can use the Divergence Theorem to find:

$$\begin{aligned}
0 &= \frac{1}{4\pi r^2} \int \int \int_{B(\mathbf{x}_0, r)} \operatorname{div}(\nabla u) dV \\
0 &= \frac{1}{4\pi r^2} \int \int \int_{B(\mathbf{x}_0, r)} \Delta u dx dy dz
\end{aligned}$$

Now for this integral to equal 0 then $\Delta u = 0$ for all $\mathbf{x}_0 \in \Omega$. To prove, Suppose $\Delta u > 0$, then by definition there would be a $B(\mathbf{x}_0, r)$ for which $\Delta u > 0$ therefore $\frac{1}{4\pi r^2} \int \int \int_{B(\mathbf{x}_0, r)} \Delta u dx dy dz > 0$ which is a contradiction as we know that expression must equal 0. Similarly, suppose $\Delta u < 0$ then by definition there would a $B(\mathbf{x}_0, r)$ for which $\Delta u < 0$ therefore $\frac{1}{4\pi r^2} \int \int \int_{B(\mathbf{x}_0, r)} \Delta u dx dy dz < 0$ which is a contradiction again. Therefore, we are justified in stating that $\Delta u = 0$ for all $\mathbf{x}_0 \in \Omega$.

Problem 4

Verify that isothermal harmonic parametrizations as defined on page 501 of [6] satisfy the minimal surface equation (3.5) on page 500 of [6].

Solution:

A parametrization is isothermal if for a function $X(r, s) = (x(r, s), y(r, s), z(r, s))$, $|X_r|^2 = |X_s|^2$ and $X_r \cdot X_s = 0$. But most importantly, a isothermal parametrization is minimal if and only if the component functions x, y , and z are harmonic functions meaning that

$x_{rr} + x_{ss} = y_{rr} + y_{ss} = z_{rr} + z_{ss} = 0$. This property of each component being harmonic implies that the Laplacian of the parametrization $\Delta X = X_{rr} + X_{ss} = 0$ which means that the mean curvature vanishes everywhere which is exactly how solutions to equation (3.5) are defined. Therefore by definition, isothermal parametrizations who have harmonic component functions are solutions to equation (3.5) meaning they are minimal surfaces.

End of Final Project "As Flat As Possible"